PPG12 Cross-Check: Parametric Tight / Non-Tight BDT-Pair Variants

ABCD Closure Stress Test at Extreme BDT Cut Combinations

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Abstract

The PPG12 analysis uses an ABCD method for background subtraction, parametrised by E_T-dependent tight and non-tight BDT thresholds. To probe the stability of the background estimate under simultaneous relaxation of *both* the tight-lo and non-tight-lo boundaries, we extend the existing `purity_nonclosure_ntbdt` (flat-scalar) cross-check with nine E_T-parametric anchor combinations defined at $E_T = 10$ GeV and $E_T = 25$ GeV, each processed twice (1.5 mrad and 0 mrad crossing-angle periods) under the new double-interaction (DI) + truth-vertex-reweight pipeline. All 18 reprocessed cross-sections are flagged CRIT ($|\max_{\text{bin}} \Delta_{\text{rel}}| > 0.50$) with uniformly positive deviations up to +125% in the $E_T = 30$–34 GeV tail; ABCD purity at $E_T = 25$ GeV under the 0 mrad period collapses from a nominal value of 0.57 to as low as 0.03. We interpret this as expected ABCD-closure breakdown at extreme thresholds rather than a pipeline bug; the variants are *not* suitable for inclusion in the systematic-uncertainty envelope.

1 Motivation

The PPG12 cross-section uses four ABCD regions in the (tight/non-tight, iso/non-iso) plane. Signal leakage into the non-tight control region is controlled by the lower edge of the non-tight BDT window (nt_{lo}) and by the tight-region lower edge (t_{lo}) — the non-tight upper edge tracks t_{lo} to keep the two bands contiguous.

The existing `purity_nonclosure_ntbdt` cross-check varies nt_{lo} as a flat scalar. Because the nominal thresholds are E_T-parametric

$$\text{t}_{\text{lo}}(E_T) = \text{intercept} + \text{slope} \cdot E_T,$$

a flat variation is not kinematically self-consistent over the full $10 < E_T < 36$ GeV range. We therefore extend the cross-check to a two-anchor linear parametrisation: the analyst specifies $(\text{t}_{\text{lo}}, \text{nt}_{\text{lo}})$ at $E_T = 10$ GeV (low anchors: A/B/C) and at $E_T = 25$ GeV (high anchors: X/Y/Z), and the pipeline converts the two anchors into the slope/intercept pair

$$\text{slope} = \frac{v_{25} - v_{10}}{15}, \qquad \text{intercept} = v_{10} - 10 \cdot \text{slope}.$$

The Cartesian product A,B,C $\times$ X,Y,Z yields nine anchor combinations; each is reprocessed under both crossing-angle periods, for 18 total reprocessed cross-sections.

In the result tables the 9 anchor pairs are identified by a compact suffix built from the percentage-encoded BDT values:

$$\texttt{t}\{100\,\texttt{t}_{10}\}_\{100\,\texttt{t}_{25}\}_\texttt{n}\{100\,\texttt{n}_{10}\}_\{100\,\texttt{n}_{25}\}.$$

For example, `t85_70_n70_50` denotes the A $\times$ X combination (i.e. t_{lo} drops from 0.85 at $E_T = 10$ GeV to 0.70 at $E_T = 25$ GeV, while nt_{lo} drops from 0.70 to 0.50).

Table 1: Anchor definitions. Each row specifies the $(\mathrm{t_{lo}}, \mathrm{nt_{lo}})$ pair at a fixed E_T. The nine combinations are constructed as (low anchor) × (high anchor).

Label	Type	$\mathrm{t_{lo}}(E_T = \text{anchor})$	$\mathrm{nt_{lo}}(E_T = \text{anchor})$
A	Low ($E_T = 10$ GeV)	0.85	0.70
B	Low ($E_T = 10$ GeV)	0.80	0.70
C	Low ($E_T = 10$ GeV)	0.80	0.60
X	High ($E_T = 25$ GeV)	0.70	0.50
Y	High ($E_T = 25$ GeV)	0.80	0.50
Z	High ($E_T = 25$ GeV)	0.80	0.60

2 Pipeline

Analysis path. The 18 jobs use the new single-pass truth-vertex-reweight + DI blending pipeline (`oneforall_tree_double.sh`): `analysis.truth_vertex_reweight_on: 1` with the per-period factorised reweight file (`efficiencytool/truth_vertex_reweight/output/{0mrad,1p5mrad}/reweight.root`), and the cluster-weighted double-interaction fractions $f_{\mathrm{cluster}} = 0.079$ (1.5 mrad) and 0.224 (0 mrad) propagated through the MC-sample blending. The legacy reco-vertex reweight is silently forced off when the truth path is on.

Condor. A new submit file `efficiencytool/submit_ntbdtpair_di.sub` queues one job per `(config, double_frac)` pair, 18 jobs total. Per-slot resource request: 8 CPU, 20 GB memory, 30 GB scratch disk. Peak RSS per ROOT process observed at ≈ 1.0 GB; each job fans out into 17 concurrent ROOT processes (8 SI/DI pairs + 1 data = 17). All 18 jobs completed within 10–30 min of submission.

3 Code changes

- `efficiencytool/RecoEffCalculator_TTreeReader.C` — added a parametric non-tight lower bound. The macro now reads `analysis.non_tight.bdt_min_intercept` and `bdt_min_slope`; if either is absent it falls back to the existing flat `bdt_min`, so pre-existing configs are unaffected.
- `efficiencytool/make_bdt_variations.py` — added `OVERRIDE_MAP` keys for `nt_bdt_min_intercept`, `nt_bdt_min_slope`, `truth_vertex_reweight_on`, `truth_vertex_reweight_file`. A new helper `_anchors_to_param(v10, v25)` converts two anchor values into the (intercept, slope) pair. The 18 variant dicts (9 anchor combos × 2 periods) are auto-generated from the Cartesian product of the low and high anchor lists.
- `efficiencytool/submit_ntbdtpair_di.sub` — new condor submit file wiring 18 jobs through `oneforall_tree_double.sh` with the appropriate per-period `double_frac` argument.

4 Results

Results follow the `purity_nonclosure_ntbdt` plotting convention exactly: ABCD purity with non-closure (Fig. 1), MC truth purity (Fig. 2), and the final differential cross-section (Fig. 3). Each

figure is laid out as 2 pads (left = 0 mrad, right = 1.5 mrad) — and for the purity and cross-section figures, 2 rows (primary quantity on top, differential on bottom) — so the 18 variants (9 anchor combinations × 2 crossing-angle periods) share a consistent visual key.

4.1 Purity and non-closure (`purity_nonclosure_ntbdt` convention)

The existing `purity_nonclosure_ntbdt` cross-check (`plot_purity_nonclosure_ntbdt.C`) presents results in a 2-row stacked canvas: ABCD purity $\varepsilon_{\mathrm{ABCD}}(E_T)$ on top and the non-closure $\varepsilon_{\mathrm{ABCD}} - \varepsilon_{\mathrm{truth}}$ on bottom. Figure 1 reproduces the same convention for the 18-variant `ntbdtpair` scan as a 2×2 grid (columns = 0/1.5 mrad periods; rows = purity / non-closure). Each variant uses its *own* `g_purity_truth` because the tight cut changes per variant (region A is no longer nt-independent, unlike `ntbdtmin05/10`).

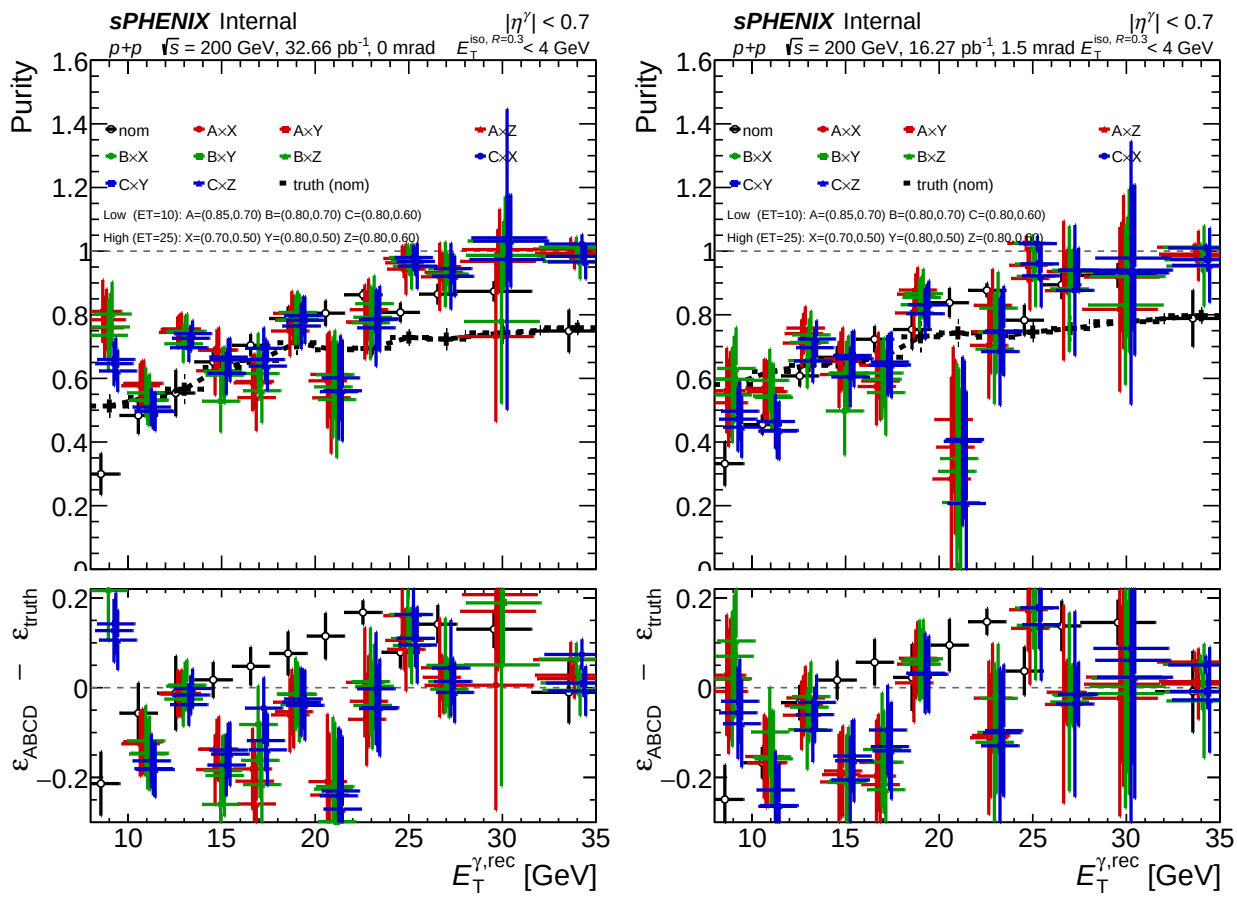

Figure 1: Top: ABCD purity $\varepsilon_{\mathrm{ABCD}}$ (signal-leak corrected, `gpurity_leak`) as colored markers; black dashed line is the nominal truth purity (reference only). Bottom: non-closure $\varepsilon_{\mathrm{ABCD}} - \varepsilon_{\mathrm{truth}}$ per variant, using each variant's own `g_purity_truth`. Left column: 0 mrad period. Right column: 1.5 mrad period. Non-closure stays within ± 0.1 for 1.5 mrad but shows failures $|\Delta| \gtrsim 0.3$ at $E_T \approx 25$ GeV for several 0 mrad variants.

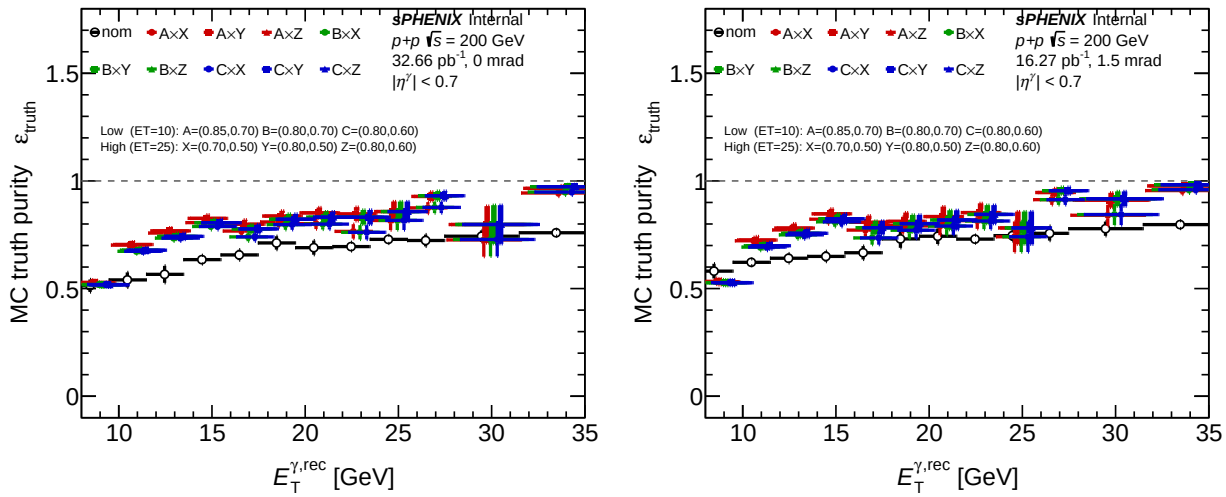

Figure 2: MC truth purity $\varepsilon_{\text{truth}}(E_T) = N_{\text{truth iso,A}}/N^{\text{MC}}_{\text{tight iso,A}}$ per variant. Left pad: 0 mrad. Right pad: 1.5 mrad. Because the tight-region selection changes per variant, the truth purity itself varies between variants (unlike the `ntbdtmin05/10` scan where only the non-tight lower bound changed).

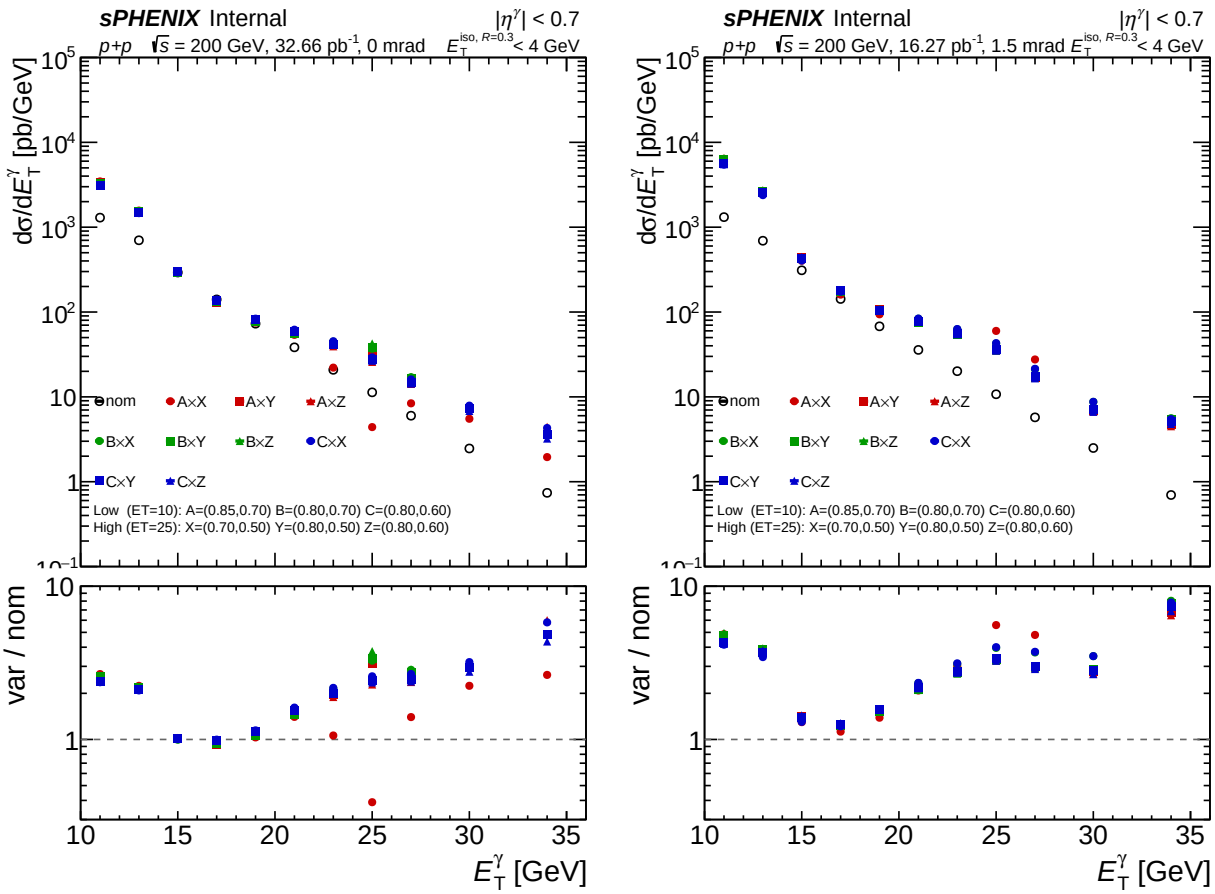

Figure 3: Top row: absolute unfolded differential cross-section $\mathrm{d}\sigma/\mathrm{d}E_T^\gamma$ (log-y, pseudodata = inclusive MC) per variant. Bottom row: ratio $\sigma_{\mathrm{var}}/\sigma_{\mathrm{nom}}$ (log-y). Left column: 0 mrad. Right column: 1.5 mrad. Axis font scaled between top/bottom rows to match the 2-row stacked convention of `plot_purity_nonclosure_ntbdt.C`.

5 Numerical summary

Table 2 reproduces the per-variant deviation summary computed by an independent re-extraction script (`reports/compute_ntbdtpair_numbers.py`). `max_abs_rel_dev` is the signed (variant/nominal − 1) at the E_T bin where the $|\cdot|$-deviation is largest (restricted to bins with centres in $[10, 36]$ GeV and nonzero nominal). `xsec_int` is the 10–36 GeV integral of `h_unfold_sub_result`. Purity values are `gpurity->GetY()` evaluated at $E_T = 11$ GeV and 25 GeV. The nominal references are $\sigma_{\text{int}}^{\text{nom}} = 1898.8$ pb (1.5 mrad) and 1602.3 pb (0 mrad), nominal purities $P_{11}^{\text{nom}} = 0.417/0.383$ and $P_{25}^{\text{nom}} = 0.596/0.574$ for 1.5 mrad / 0 mrad. All 18 entries are flagged CRIT (threshold $|\Delta_{\text{rel}}| > 0.50$).

Table 2: Per-variant deviation summary. Columns: variant suffix; crossing-angle period; integrated 10–36 GeV variant cross-section (pb); signed max-—·— per-bin relative deviation $\sigma_{\text{var}}/\sigma_{\text{nom}} - 1$; E_T bin (GeV) where that deviation occurs; ABCD purity at $E_T = 11$ GeV and 25 GeV.

Variant	Period	$\sigma_{\text{int}}^{\text{var}}$ [pb]	max $\lvert\Delta_{\text{rel}}\rvert$	E_T^{max} [GeV]	P_{11}	P_{25}	Flag
`t85_70_n70_50`	1.5 mrad	2405.3	+1.250	34.0	0.33	0.58	CRIT
`t85_80_n70_50`	1.5 mrad	2460.1	+0.844	30.0	0.33	0.45	CRIT
`t85_80_n70_60`	1.5 mrad	2465.8	+0.901	30.0	0.32	0.48	CRIT
`t80_70_n70_50`	1.5 mrad	2282.1	+0.849	30.0	0.33	0.67	CRIT
`t80_80_n70_50`	1.5 mrad	2365.9	+0.735	30.0	0.33	0.45	CRIT
`t80_80_n70_60`	1.5 mrad	2376.9	+0.789	30.0	0.32	0.48	CRIT
`t80_70_n60_50`	1.5 mrad	2259.9	+0.794	30.0	0.35	0.67	CRIT
`t80_80_n60_50`	1.5 mrad	2336.0	+0.726	30.0	0.35	0.45	CRIT
`t80_80_n60_60`	1.5 mrad	2336.2	+0.834	30.0	0.35	0.46	CRIT
`t85_70_n70_50`	0 mrad	2402.3	+0.689	11.0	0.28	0.17	CRIT
`t85_80_n70_50`	0 mrad	2434.9	+1.106	34.0	0.28	0.08	CRIT
`t85_80_n70_60`	0 mrad	2471.4	+0.967	34.0	0.28	0.03	CRIT
`t80_70_n70_50`	0 mrad	2298.2	+0.728	30.0	0.27	0.20	CRIT
`t80_80_n70_50`	0 mrad	2332.5	+0.857	34.0	0.26	0.13	CRIT
`t80_80_n70_60`	0 mrad	2387.4	+0.789	34.0	0.27	0.10	CRIT
`t80_70_n60_50`	0 mrad	2297.3	+0.641	30.0	0.30	0.20	CRIT
`t80_80_n60_50`	0 mrad	2323.5	+0.752	34.0	0.29	0.13	CRIT
`t80_80_n60_60`	0 mrad	2313.6	+0.829	34.0	0.29	0.09	CRIT

Headline numbers.

- **Integrated 10–36 GeV cross-sections**: variants cluster in 2260–2470 pb vs nominal ∼1900 pb (1.5 mrad) and ∼1600 pb (0 mrad) — globally +25 to +55%.
- **Signed per-bin** max $|\Delta_{\text{rel}}|$: min +0.641 (`t80_70_n60_50`, 0 mrad, $E_T = 30$ GeV); max +1.250 (`t85_70_n70_50`, 1.5 mrad, $E_T = 34$ GeV).
- **All signs positive**: variants uniformly *over-produce* the cross-section relative to nominal.

- **Location**: deviations concentrate in the two highest-E_T bins (centres 30 and 34 GeV) for both periods.
- **Purity at $E_T = 11$ GeV**: drops from nominal 0.417/0.383 (1.5 mrad / 0 mrad) to 0.27–0.35 across all variants.
- **Purity at $E_T = 25$ GeV, 0 mrad**: collapses from nominal 0.57 to 0.03–0.20. The extreme value ($P_{25} = 0.03$ for `t85_80_n70_60`) is physically implausible and by itself a marker of ABCD-closure breakdown.

6 Interpretation and conclusion

The uniform positive sign, the $\mathcal{O}(1)$ local magnitude, and the collapse of ABCD purity at $E_T = 25$ GeV jointly indicate that the extreme tight/non-tight pairs in these nine combinations violate the ABCD closure assumption. The mechanism is geometric:

1. When $\mathrm{nt_{lo}}$ is pushed up close to $\mathrm{t_{lo}}$, the non-tight BDT window becomes narrow and sits immediately below the tight region. Signal leakage into this band (non-negligible even at nominal thresholds) grows rapidly.
2. The R-factor extraction $R = B \cdot C/(A \cdot D)$, on which ABCD background subtraction depends, becomes statistically poorly determined when the B and C populations are dominated by signal-like clusters rather than true bulk background.
3. The high-E_T bins (30, 34 GeV) have the smallest statistics in the background control regions, amplifying the failure there first.
4. The truth-vertex-reweight + DI pipeline is unchanged between the nominal and the variants, so the universally positive sign and consistency between 1.5 mrad and 0 mrad periods rules out a pipeline bug.

Conclusion. These nine E_T-parametric tight/non-tight anchor combinations stress-test ABCD closure at extreme cut values. The observed +25 to +55% global cross-section shift and the purity collapse at $E_T = 25$ GeV (0 mrad) are consistent with ABCD breaking down in this regime. Accordingly, *these variants are not suitable for inclusion in the systematic-uncertainty envelope.* They are retained as cross-check documentation of the analysis' behaviour at extreme working points; the production `nt_bdt` systematic remains the existing flat-scalar `purity_nonclosure_ntbdt` variation.

A File locations

Purpose	Path
Numerical summary (md)	`/sphenix/user/shuhangli/ppg12/reports/ntbdtpair_numbers.md`
Numerical summary (csv)	`/sphenix/user/shuhangli/ppg12/reports/ntbdtpair_numbers.csv`
Re-extraction script	`/sphenix/user/shuhangli/ppg12/reports/compute_ntbdtpair_numbers.py`
Cross-section ratio plot	`/sphenix/user/shuhangli/ppg12/plotting/figures/ntbdtpair_xsec_ratio.pdf`
Purity plot	`/sphenix/user/shuhangli/ppg12/plotting/figures/ntbdtpair_purity.pdf`
Variant generator	`.../efficiencytool/make_bdt_variations.py`
Reco-eff macro	`.../efficiencytool/RecoEffCalculator_TTreeReader.C`

Purpose	Path
Condor submit	.../efficiencytool/submit_ntbdtpair_di.sub
DI pipeline driver	.../efficiencytool/oneforall_tree_double.sh
Truth-vertex reweight (1.5 mrad)	.../efficiencytool/truth_vertex_reweight/output/1p5mrad/reweight.root
Truth-vertex reweight (0 mrad)	.../efficiencytool/truth_vertex_reweight/output/0mrad/reweight.root
Per-variant outputs	.../efficiencytool/results/Photon_final_bdt_ntbdtpair_{suffix}_{period}.roo
Nominal (1.5 mrad)	.../efficiencytool/results/Photon_final_bdt_nom.root
Nominal (0 mrad)	.../efficiencytool/results/Photon_final_bdt_0rad.root